



FYP Regression Algorithms For Learning

By: Syed Maesum Hussain Zaidi

Introduction & Context

Regression Algorithms for Learning & Its Application on Cryptocurrency Market

- Main Concepts
- Progress Report
- Next Steps
- Summary



**Question
to think
about
during the
presentation?**



Main Concepts

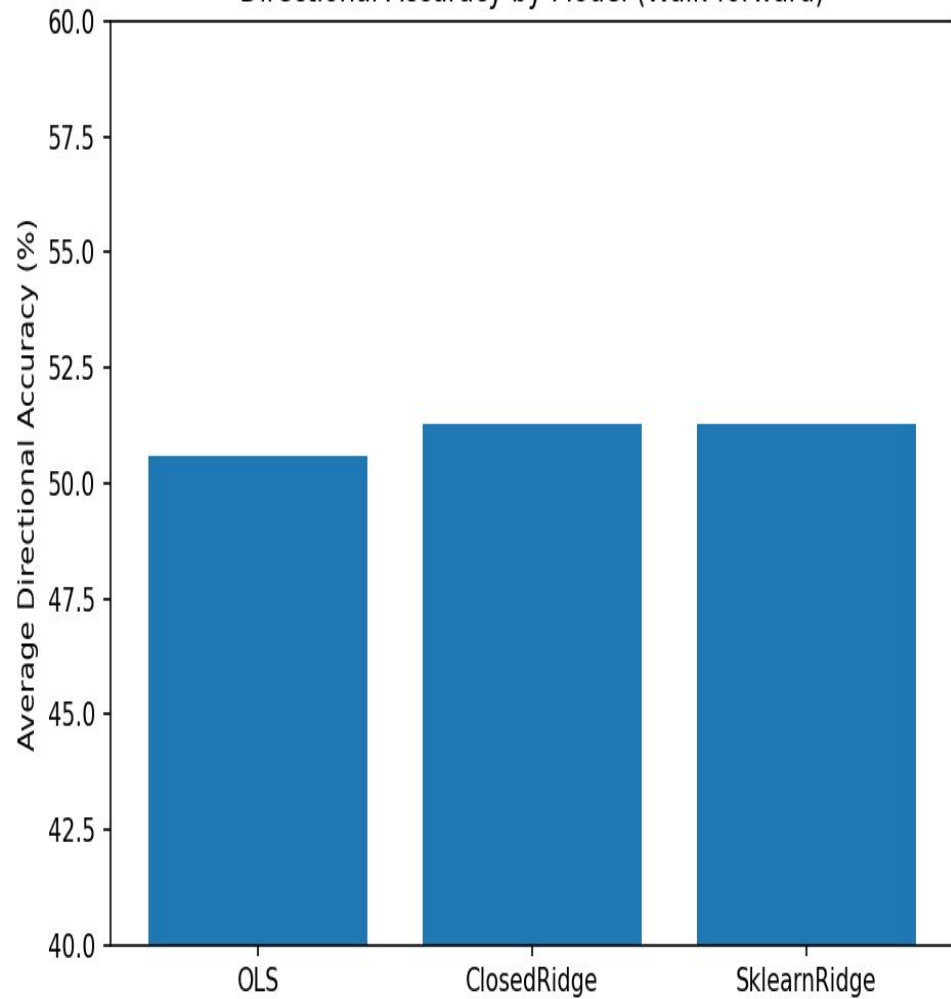
- Regression Algorithms Implementation
- Real World Dataset Preprocessing (Bitcoin-US Dollar)
- Future Price Prediction
- Regression Algorithms Performance Comparison
- Different Learning Algorithm Implementation
- Overall Comparison between Different Implementations



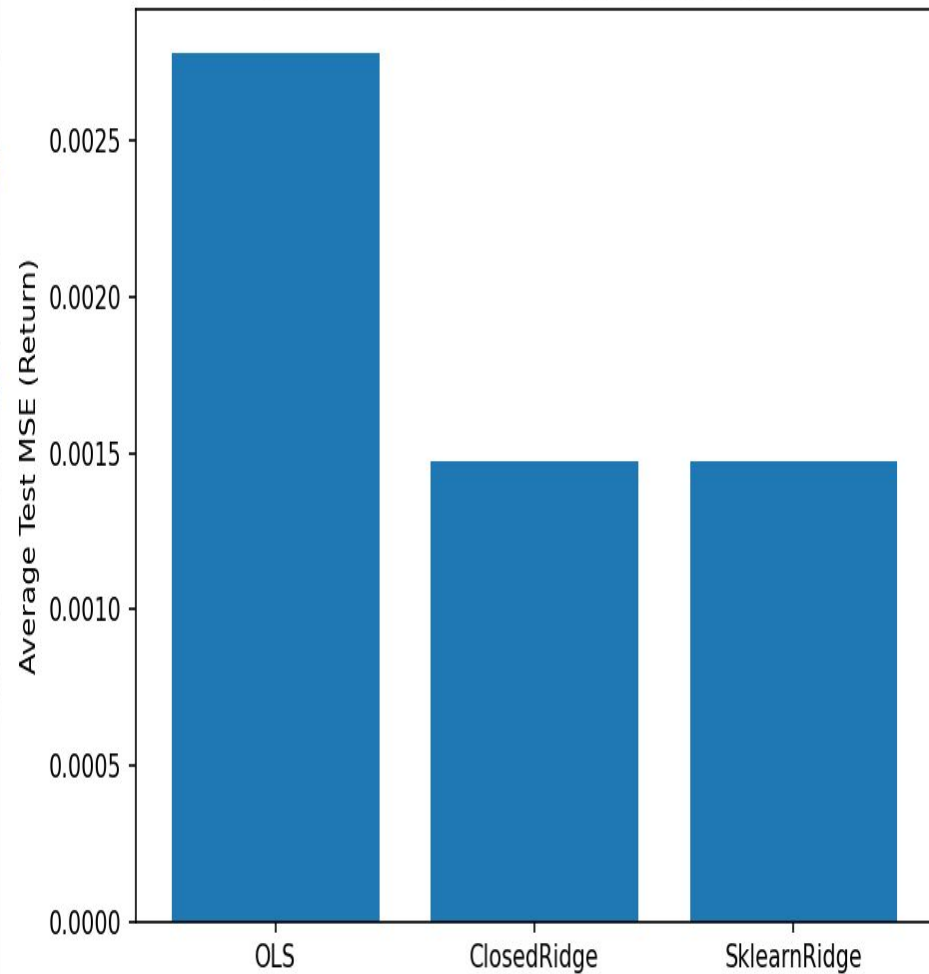
Progress Report

- Report 1: Mathematical Foundations of Regression Algorithms
- Report 2: Short Literature Review on works done on related topics
- Proof of Concept program on a Synthetic Dataset
- Report 3: Connecting the previous reports with the Proof of Concept Program
- Program applying Ridge and OLS Regression on BTC-USD dataset to give price prediction for the next day.

Directional Accuracy by Model (Walk-forward)



Average Return MSE by Model (Walk-forward)



Next Steps

- Implement Lasso Regression
- Implement Nearest Neighbour Algorithm
- Implement Neural Networks Algorithm
- Implement a Graphical User Interface
- Write Analytical reports based on the Performance comparison from different Learning Algorithms.



Conclusion

- Implementing Different Learning Algorithms
- For BTC-USD Price Prediction
- Current Progress: Ridge and OLS Regression
- Next Steps: Lasso Regression, Nearest Neighbour & Neural Networks
- Goal: Compare the performances of Different Learning Algorithms



Thank You